Problem 2.1: Non-overlapping wave functions

Consider a particle and two normalized energy eigenfunctions $\psi_1(\mathbf{x})$ and $\psi_2(\mathbf{x})$ corresponding to the eigenvalues $E_1 \neq E_2$. Assume that the eigenfunctions vanish outside the two non-overlapping regions Ω_1 and Ω_2 respectively.

(a) Show that, if the particle is initially in region Ω_1 then it will stay there forever.

(b) If, initially, the particle is in the state with wave function

$$\psi(\mathbf{x}, 0) = \frac{1}{\sqrt{2}}[\psi_1(\mathbf{x}) + \psi_2(\mathbf{x})],$$

show that the probability density $\rho = |\psi(\mathbf{x}, t)|^2$ is independent of time.

(c) Now assume that the two regions Ω_1 and Ω_2 overlap partially. Starting with the initial wave function of case (b), show that the probability density ρ is a periodic function of time.

(d) Starting with the same initial wave function and assuming that the two eigenfunctions are real and isotropic, take the two partially overlapping regions Ω_1 and Ω_2 to be two concentric spheres of radii $R_1 > R_2$. Compute the probability current j that flows through Ω_1.

Hint: The probability current density is given by

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi}\left(\psi^*(\mathbf{x}, t)\nabla\psi(\mathbf{x}, t) - \psi(\mathbf{x}, t)\nabla\psi^*(\mathbf{x}, t)\right).$$

This problem can be solved in many ways. You can choose however you want to approach it. Below, I provide four different solutions using four different formalisms: wavefunctions, Dirac notation, propagators, and density matrices.